

MAT 344 HW06

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Problem 1 (p.161#1ce). Find the poles and residues of the following functions: (c) $\frac{1}{\sin z}$; (e) $\frac{1}{\sin^2 z}$.

Solution. The zeroes of $\sin z$ (hence also of $\sin^2 z$) are the points at which

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i} = 0 \iff e^{2iz} = 1 \iff z = n\pi \text{ for } n \in \mathbb{Z}.$$

At $z = n\pi$, $\sin z$ locally has the power series representation

$$\sum_{k=1}^{\infty} \frac{(z - n\pi)^{2k-1}}{(2k-1)!},$$

so

$$1 = \lim_{z \rightarrow n\pi} \frac{z - n\pi}{\sin(z)} = \lim_{z \rightarrow n\pi} \frac{(z - n\pi)^2}{\sin^2(z)}$$

(with the second equality coming from the continuity of squaring). This means that for every integer n , $n\pi$ is an order 1 pole of $\frac{1}{\sin z}$ and an order 2 pole of $\frac{1}{\sin^2 z}$. The formula for residues that we calculated in class implies that the answer to (c) is

$$\text{Res}_{z=n\pi} \left(\frac{1}{\sin z} \right) = \lim_{z \rightarrow n\pi} \frac{z - n\pi}{\sin z} = \boxed{1}.$$

For (e),

$$\begin{aligned} \text{Res}_{z=n\pi} \left(\frac{1}{\sin^2 z} \right) &= \lim_{z \rightarrow n\pi} \frac{d}{dz} \frac{(z - n\pi)^2}{\sin^2 z} \\ &= \lim_{z \rightarrow 0} \frac{2z \sin^2 z - 2z^2 \sin z \cos z}{\sin^4 z} \\ &= \lim_{z \rightarrow 0} \frac{2z \sin z - 2z^2 \cos z}{\sin^3 z} \\ &= \lim_{z \rightarrow 0} \frac{2 \sin z - 2z \cos z}{z^2}. \end{aligned}$$

If we write out the power series for $\sin z$ and $z \cos z$, the linear terms cancel, so that $\sin z - z \cos z$ has a zero of order 3 at 0, meaning that the limit at the end, hence the residues of $\frac{1}{\sin^2 z}$, are $\boxed{0}$. $\square$

Problem 2 (p.161#cd). Evaluate the following integrals by the method of residues:

$$(c) \int_{-\infty}^{\infty} \frac{x^2 - x + 2}{x^4 + 10x^2 + 9} dx, \quad (d) \int_0^{\infty} \frac{x^2}{(x^2 + a^2)^3} dx, \quad a \in \mathbb{R}.$$

Solution. (c) Let γ_R be the counterclockwise-oriented semicircular arc above the real axis of radius R centered at 0. By the residue theorem, if R is large enough (greater than 3 should suffice, since the roots of $x^4 + 10x^2 + 9$ are $\pm i, \pm 3i$), then

$$\int_{-R}^R \frac{x^2 - x + 2}{x^4 + 10x^2 + 9} dx + \int_{\gamma_R} \frac{x^2 - x + 2}{x^4 + 10x^2 + 9} dx = 2\pi i \left(\text{Res}_i \left(\frac{x^2 - x + 2}{x^4 + 10x^2 + 9} \right) + \text{Res}_{3i} \left(\frac{x^2 - x + 2}{x^4 + 10x^2 + 9} \right) \right).$$

Noting that the roots of the denominator each have multiplicity 1, i.e. are simple poles, the sum of the residues equals

$$\frac{i^2 - i + 2}{(i + i)(i^2 + 9)} + \frac{(3i)^2 - 3i + 2}{((3i)^2 + 1)(3i + 3i)} = \frac{1 - i}{16i} + \frac{-7 - 3i}{-48i} = \frac{1}{48i}(3 - 3i + 7 + 3i) = \frac{5}{24i}.$$

Taking R arbitrarily large, the latter integral on the right-hand side goes to zero because

$$\int_{\gamma_R} \frac{x^2 - x + 2}{x^4 + 10x^2 + 9} dx = \int_0^{\pi} \frac{R^2 e^{2i\theta} - R e^{i\theta} + 2}{R^4 e^{4i\theta} + 10R^2 e^{2i\theta} + 9} \cdot R e^{i\theta} d\theta = \int_0^{\pi} \frac{1}{R} \cdot \frac{e^{2i\theta} - R^{-1} e^{i\theta} + 1R^{-2}}{e^{4i\theta} + 10R^{-2} e^{2i\theta} + 9} \cdot e^{i\theta} d\theta,$$

which, in magnitude, is bounded above by

$$\int_0^{\pi} \frac{1}{R} \cdot \frac{1 + R^{-1} + 2R^{-2}}{1 - 10R^{-1} - 9R^{-2}} d\theta = \frac{\pi}{R} \cdot O(1).$$

Thus, in the limit, we get that

$$\int_{-\infty}^{\infty} \frac{x^2 - x + 2}{x^4 + 10x^2 + 9} dx = 2\pi i \cdot \frac{5}{24i} = \boxed{\frac{5\pi}{12}}.$$

(d) Since the integrand is even,

$$\int_0^{\infty} \frac{x^2}{(x^2 + a^2)^3} dx = \frac{1}{2} \int_{-\infty}^{\infty} \frac{x^2}{(x^2 + a^2)^3} dx.$$

By the same procedure as with (c),

$$\int_{-\infty}^{\infty} \frac{x^2}{(x^2 + a^2)^3} dx = 2\pi i \text{Res}_{ai} \left(\frac{x^2}{(x^2 + a^2)^3} \right).$$

Since $(x^2 + a^2)^3 = (x + ai)^3(x - ai)^3$, ai is a pole of order 3, so the residue is equal to

$$\frac{1}{2!} \cdot \frac{d^2}{dx^2} \frac{x^2(x - ai)^3}{(x^2 + a^2)^3} \Big|_{x=ai} = \frac{1}{2} \left(\frac{12(ai)^2}{(2ai)^5} - \frac{12ai}{(2ai)^4} + \frac{2}{(2ai)^3} \right) = \frac{-i}{16a^3},$$

meaning

$$\int_0^{\infty} \frac{x^2}{(x^2 + a^2)^3} \frac{1}{2} \int_{-\infty}^{\infty} \frac{x^2}{(x^2 + a^2)^3} dx = \frac{1}{2} \cdot 2\pi i \cdot \frac{-i}{16a^3} = \boxed{\frac{\pi}{16a^3}}. \quad \square$$

Problem 3 (p.136#1). Show that $|f(z)| \leq 1$ for $|z| \leq 1$ implies

$$\frac{|f'(z)|}{1 - |f(z)|^2} \leq \frac{1}{1 - |z|^2}.$$

Solution. Using equation (36) on p.136 with $R = M = 1$, we get that for any z_0 in the unit disk,

$$\left| \frac{(f(z) - f(z_0))/(z - z_0)}{1 - \overline{f(z_0)}f(z)} \right| \leq \left| \frac{1}{1 - \overline{z_0}z} \right|.$$

Taking the limit as $z \rightarrow z_0$ gives the desired inequality at z_0 , since the denominators become $1 - \overline{f(z_0)}f(z_0)$ and $1 - \overline{z_0}z_0$ and the numerator on the left becomes $f'(z_0)$. $\square$

Problem 4 (p.136#5). Show by use of Schwarz's lemma that every one-to-one conformal mapping of a disk onto another (or a half-plane) is given by a linear transformation.

Solution. Suppose first that f sends the unit disk to itself biholomorphically. Then, if $a \in \Delta$ is the point such that $f(a) = 0$, compose f with the linear transformation

$$T: \Delta \rightarrow \Delta, \quad Tz = \frac{a - z}{1 - \overline{a}z}$$

sending 0 to a , so that $g = f \circ T$ is an automorphism of Δ sending 0 to 0. Then let $h: \Delta \rightarrow \Delta$ be the inverse of g , and apply Schwarz's lemma to g and h to get that $|g'(0)|$ and $|h'(0)|$ are both at most 1. By the chain rule $1 = g'(0) \cdot h'(g(0)) = g'(0) \cdot h'(0)$, implying that $|g'(0)| = |h'(0)| = 1$. The equality case of Schwarz's lemma tells us that $g(z) = cz$ for some constant c with $|c| = 1$, hence $f(z) = cT^{-1}(z) = cT(z)$, which is a linear transformation.

In the general case, where $f: U \rightarrow V$ is biholomorphic, the function $\varphi \circ f \circ \psi^{-1}: \Delta \rightarrow \Delta$ where $\psi: U \rightarrow \Delta$ and $\varphi: V \rightarrow \Delta$ are linear transformations satisfies the hypotheses of the problem, so the previous paragraph tells us that it is a linear transformation, hence so is f . $\square$

Problem 5 (p.136#6). If γ is a piecewise differentiable arc contained in $|z| < 1$ the integral

$$\int_{\gamma} \frac{|dz|}{1 - |z|^2}$$

is called the *noneuclidean* or *hyperbolic* length of γ . Show that an analytic function $f(z)$ with $|f(z)| < 1$ for $|z| < 1$ maps every γ to an arc with smaller or equal noneuclidean length. Deduce that a linear transformation of a disk onto itself preserves noneuclidean lengths, and check the result by an explicit computation.

Solution. Parametrize γ as a function $\gamma: [0, 1] \rightarrow \Delta$. By p.136#1, we have

$$\frac{|\gamma'(t) \cdot f'(\gamma(t))|}{1 - |f(\gamma(t))|^2} \leq \frac{|\gamma'(t)|}{1 - |\gamma(t)|^2}$$

for all $t \in [0, 1]$, so integrating over $[0, 1]$ and doing a change of variables gives

$$\int_{f(\gamma)} \frac{|dz|}{1 - |z|^2} \leq \int_{\gamma} \frac{|dz|}{1 - |z|^2}.$$

If T is a linear transformation on the unit disk then the hyperbolic length of $T\gamma$ is less than or equal to that of γ and the hyperbolic length of $T\gamma$ is less than or equal to that of $T^{-1}(T\gamma) = \gamma$, i.e. $T\gamma$ and γ have the same hyperbolic length.

Concretely, suppose that $f(z) = \frac{c(a-z)}{1-\bar{a}z}$ is an automorphism of Δ , where $|c| = 1$ and $|a| < 1$. Then for a piecewise differentiable curve $\gamma: [0, 1] \rightarrow \mathbb{C}$,

$$\frac{|\gamma'(t) \cdot f'(\gamma(t))|}{1 - |f(\gamma(t))|^2} = \frac{|\gamma'(t)| \cdot \left| \frac{1-|a|^2}{(1-\bar{a}\gamma(t))^2} \right|}{1 - \left| \frac{a-\gamma(t)}{1-\bar{a}\gamma(t)} \right|^2} = \frac{|\gamma'(t)| \cdot (1 - |a|^2)}{|1 - \bar{a}\gamma(t)|^2 - |a - \gamma(t)|^2}.$$

The squared magnitudes in the denominator can be expanded by writing them as the argument times its conjugate to get

$$(1 - \bar{a}\gamma(t))(1 - a\overline{\gamma(t)}) - (a - \gamma(t))(\bar{a} - \overline{\gamma(t)}) = 1 - |a|^2 - |\gamma(t)|^2 + |a|^2 |\gamma(t)|^2 = (1 - |a|^2)(1 - |\gamma(t)|^2)$$

so that the fraction above equals

$$\frac{|\gamma'(t) \cdot f'(\gamma(t))|}{1 - |f(\gamma(t))|^2} = \frac{|\gamma'(t)|}{1 - |\gamma(t)|^2}.$$

Thus,

$$\int_{\gamma} \frac{|dz|}{1 - |z|^2} = \int_0^1 \frac{|\gamma'(t)|}{1 - |\gamma(t)|^2} dt = \int_0^1 \frac{|\gamma'(t) \cdot f'(\gamma(t))|}{1 - |f(\gamma(t))|^2} dt = \int_{f(\gamma)} \frac{|dz|}{1 - |z|^2}$$

as desired. □

Problem 6 (p.186#1). Prove that the Laurent development is unique.

Solution. This follows from the fact that the coefficients a_n in the Laurent series

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

can be directly computed in terms of f , z_0 , and n as

$$a_n = \frac{1}{2\pi i} \int_{|z-z_0|=r} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta,$$

where r is between the smaller and larger radius of the annulus of convergence of the series. Note also that the integral is independent of the particular choice of r by homotopy invariance/Green's theorem. □